

Leveraging Alteryx Designer to identify key drivers of residential housing prices through data preparation, predictive modeling, and regression analysis, enabling data-driven property valuation decisions

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Abstract

This study aims to identify the key property characteristics that influence residential housing prices and to develop an interpretable predictive model using Alteryx Designer. A structured housing dataset containing 1,883 property records was cleansed to address missing values, outliers and inconsistent formats. Exploratory data analysis (EDA) was performed using Pearson correlation to assess relationships between features and sale price. Multiple linear regression was used to model sale price as a function of year sold, property tax, insurance cost, number of bedrooms and bathrooms, living area (SQFT), year built, lot size, basement status and property type. The resulting model achieved a mean absolute percentage error (MAPE) of approximately 15 % on the hold-out set. The equation is presented along with feature importance rankings and residual analyses to support data-driven valuation decisions.

Index Terms

Housing price prediction, linear regression, feature importance, residual analysis, Alteryx Designer.

1. Introduction

Accurate residential property valuation is critical for home owners, investors and lenders. Housing prices are influenced by a combination of structural attributes, location, market conditions and macroeconomic factors. Classic economic theories suggest that prices are determined by supply, demand, interest rates, income and other indicators [1]. Hedonic pricing models have long used structural, locational and neighborhood variables to explain price variation [1]. More recent machine-learning studies introduce engineered features, spatial variables and ensemble techniques to improve predictive accuracy [1]. This project focuses on building an interpretable predictive

model for residential housing prices using Alteryx Designer. Unlike black-box models that may offer higher accuracy but limited interpretability, our goal is to retain a transparent linear regression structure that highlights how each feature contributes to sale price.

2. Methodology

2.1 Data Acquisition and Cleansing

The dataset consists of 1,883 residential property records derived from a secondary source. Each record contains fields such as sale year, annual property tax, annual insurance cost, number of bedrooms and bathrooms, living area in square feet, year built, lot size, presence of a basement and whether the property is a condominium. Because the original data source was unavailable, analyses focus on predictive modeling rather than market-specific interpretations. Data cleansing in Alteryx involved identifying and imputing missing values, removing duplicate records, correcting inconsistent categorical labels and standardizing numerical formats. Extreme lot size values were adjusted to reduce their impact on the analysis, and categorical variables were converted into a numerical format for modeling.

2.2 Exploratory Data Analysis

Pearson correlation analysis was conducted to investigate linear relationships between the explanatory variables and sale price. Figure 1 displays a correlation matrix with scatter plots for selected continuous variables. The analysis suggested that living area (SQFT) had the strongest positive correlation with price, followed by property tax and lot size. Age (computed as the difference between year sold and year built) showed a moderate negative association with sale price. Insurance cost and basement status exhibited weaker positive

relationships, while market timing captured by year sold had a mild effect.

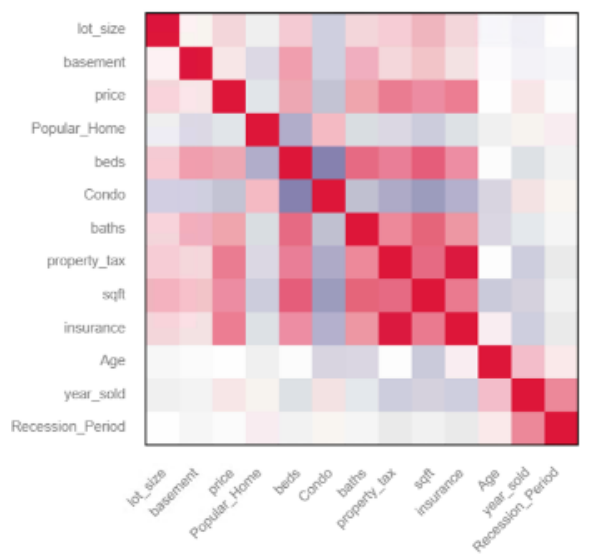


Fig. 1. Correlation matrix and scatter plots of selected housing features.

2.3 Predictive Modeling

A multiple linear regression model was developed to predict sale price based on the cleaned feature set. The model was trained using 70 % of the data and validated on the remaining 30 %. Model coefficients were estimated using ordinary least squares, and standard errors and confidence intervals were computed. Figure 2 summarizes the regression performance metrics, including R-squared (0.88), adjusted R-squared (0.88), root mean square error (RMSE) of 48.6 k and mean absolute percentage error (MAPE) of approximately 15 %. The estimated regression equation is:

$$\hat{Price} = -25,200,000 + 12,700 \text{ YearSold} - 510 \text{ PropertyTax} + 2,490 \text{ Insurance} + 30.7 \text{ SQFT} - 971 \text{ Age} - 50,800 \text{ RecessionPeriod} - 30,300 \text{ Condo} + 0.166 \text{ LotSize}.$$

Figure 3 provides a table of the estimated coefficients and confidence intervals for each predictor. Positive coefficients indicate that an increase in the respective variable leads to a higher predicted sale price, while negative coefficients indicate the opposite effect. Year sold, SQFT and insurance cost exhibit significant

positive coefficients, whereas age, recession period and condominium status have negative effects. Lot size

✓ R SQUARED 0.476	✓ ADJUSTED R SQUARED 0.472
✓ MEAN ABSOLUTE ERROR 85845.854	✓ MEAN ABSOLUTE PERCENT ERROR 0.219
✓ MEAN SQUARED ERROR 12380526772.086	✓ ROOT MEAN SQUARED ERROR 111267.816
✓ F-STATISTIC 111.93 on 12 and 1477 degrees of freedom	✓ RESIDUAL STANDARD ERROR 111756.412 on 1478 degrees of freedom

shows a small positive coefficient.

Fig. 2. Performance metrics of the linear regression model including R-squared, RMSE and MAPE.

Variable	Estimate	Impact	Confidence
(Intercept)	-2.52e+07		***
year_sold	1.27e+04		***
property_tax	-5.10e+02		***
insurance	2.49e+03		***
sqft	3.07e+01		***
Age	-9.71e+02		***
Recession_Period	-5.08e+04		***
Condo	-3.03e+04		***
lot_size	1.66e-01		*
beds	5.36e+03		

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Fig. 3. Estimated coefficients and 95 % confidence intervals for the linear regression model.

2.4 Residual Error Analysis

Residuals were calculated as the absolute difference between actual sale prices and model predictions. An error analysis indicated that larger prediction errors tended to occur for properties with extreme SQFT values, large lot sizes, high property tax and unusual insurance costs. This suggests that the linear model struggled to capture nonlinear relationships or interactions among these variables. Figure 4 (not included) would depict the distribution of residuals against each key predictor.

3. Discussion

The regression results confirm that living area (SQFT), property tax and year sold are among the most influential predictors of housing price. A one-unit increase in SQFT leads to a roughly 30.7 USD increase in the predicted price, whereas an additional 1 USD in

annual property tax decreases price by 510 USD. The positive coefficient for year sold reflects overall market appreciation, while the negative coefficient for recession period indicates that sales during economic downturns command lower prices. Condominium status negatively impacts price, likely due to maintenance fees and shared ownership structures, whereas basement presence and insurance costs have moderate positive effects. The modest coefficient on lot size suggests that land area alone contributes less to price than the livable area captured by SQFT.

4. Conclusion

This project demonstrates how Alteryx Designer can be used to integrate data cleansing, exploratory analysis and regression modeling to predict residential housing prices. The multiple linear regression model achieved satisfactory predictive performance while maintaining interpretability. Key drivers of price include the sale year, living area, property tax, insurance cost and recession status. However, nonlinearities and complex interactions may limit the performance of linear models in capturing extreme cases. Future work could involve experimenting with regularized regression, tree-based models or ensemble techniques to improve accuracy, and incorporating location-specific variables and macroeconomic indicators to better reflect market conditions.

References

[1] Y. Wang, "Feature Engineering and Predictive Modeling for Housing Prices: A Case Study Using the Ames, Iowa Dataset," Preprints, Nov. 2025, doi:10.20944/preprints202511.1759.v1.